

**DOW™ SP-1500****Resin**

Description			
Product Description	DOW™ SP-1500 resin is a purge compound resin designed to use after processing of ionomer and/or acid copolymer in extrusion and coextrusion coating equipment.		
Restrictions			
Material Status	Commercial: Active		
Other Restrictions	Available in the Asia-Pacific region only.		
Typical Characteristics			
Composition	Proprietary		
Features	Compound tailored for purging of ionomer and/or acid copolymers from the typical flow passages of extrusion coating machinery.		
Characteristics / Benefits	Shortens the purging time versus using only PE behind runs of ionomer and/or acid copolymers. Lower odor and fuming compared to conventional purge compounds developed for other applications.		
Typical Properties			
Physical	Nominal Values		Test Method(s)
*Density ()	1.05 g/cm³	ASTM D792	ISO 1183
*Melt Flow Rate (190°C/2.16kg)	0.8 g/10 min	ASTM D1238	ISO 1133
Thermal	Nominal Values		Test Method(s)
*Melting Point(DSC)	107 °C (224.6 °F)	ASTM D3418	ISO 3146
Freezing Point(DSC)	95 °C (203 °F)	ASTM D3418	ISO 3146
Vicat Softening Point ()	87 °C (188.6 °F)	ASTM D1525	ISO 306
Processing Information			
*Maximum Processing Temperature	315 °C (599 °F)		
General Processing Information	*** DOW™ SP-1500 resin has a very specific procedure for proper usage. See more information below. This procedure specifies how to properly utilize the resin, and in what proper quantities for a given size extrusion system. Improper useage of the material will lessen its effectiveness, and possibly make it more challenging to remove from the system using a polyethylene resin with the appropriate melt flow rate. The maximum recommended temperature for use in purging extrusion equipment is 315°C (599°F). This is due to the foaming nature of the compound, and the increased chance of polymer expansion from the die exit as a hazard.		
Extrusion Coating/Lamination Processing	Nominal Values		
Processing Information	For an approximate 90mm size extruder: 1) Empty the extrusion coating line hopper of the ionomer or acid copolymer. 2) Introduce 25 kgs of DOW™ SP-1500 to the hopper. 3) Change the extruder temperature profile from ionomer or acid copolymer settings to the temperature profile as shown below. 4) Open the die deckles to full width, and remove the deckle rod (if pssible). 5) Raise the screw speed to the maximum as allowable by motor load (amps or torque) or pressure. 6) When DOW™ SP-1500 is exiting the die at full width, them reduce screw speed to a low value such as 10 RPM. 7) Once the extruder temperatures are level matching the below temperature profile, raise the screw RPM back to a high level as allowed by the motor load or pressure. 8) Run all 25 kgs of DOW™SP-1500, then introduce 50 kgs LDPE of a 3 to 4 dg/min Melt Index type. 9) Maintain the below temperature profile.		

- 10) While running the LDPE, move the die deckles in and out several times to loosen and flush material in those regions.
 11) Use the "Disco" process of raising and lowering extruder speeds up and down in 1 minute intervals. (See specific Disco Purge Resin Transition procedure, available from your Dow representative.)
 12) When the 50 kgs of LDPE is close to running out, open the die deckles fully.
 13) Adjust temperature profile for the next resin to be processed and then introduce the next resin as appropriate when the system is again stable.

Feed Zone	195 °C (383 °F)
Second Zone	265 °C (509 °F)
Third Zone	315 °C (599 °F)
Fourth Zone	315 °C (599 °F)
Fifth Zone	315 °C (599 °F)
Adapter Zone	315 °C (599 °F)
Die Zone	315 °C (599 °F)

FDA Status Information	DOW™ SP-1500 resin does not comply with FDA regulations permitting contact with food. This product is designed to be used to clean extruders and not for manufacture of food contact materials. A food grade LDPE should be used to purge out the DOW™ SP-1500 as described in the procedure, if the extrusion machine is being used to process food packaging materials.
Regulatory Information	For information on regulatory compliance outside of the U.S.A., consult your local Dow representative.
Safety & Handling	For information on appropriate Handling & Storage of this polymeric resin, please refer to the material Safety Data Sheet. A Product Safety Bulletin, material Safety Data Sheet, and/or more detailed information on extrusion processing and/or compounding of this polymeric resin for specific applications are available from your Dow representative.

Product Stewardship

The Dow Chemical Company and its subsidiaries (“Dow”) has a fundamental concern for all who make, distribute, and use its products, and for the environment in which we live. This concern is the basis for our Product Stewardship philosophy by which we assess the safety, health, and environmental information on our products and then take appropriate steps to protect employee and public health and our environment. The success of our Product Stewardship program rests with each and every individual involved with Dow products – from the initial concept and research, to manufacture, use, sale, disposal, and recycle of each product.

Customer Notice

Dow strongly encourages its customers to review both their manufacturing processes and their applications of Dow products from the standpoint of human health and environmental quality to ensure that Dow products are not used in ways for which they are not intended or tested. Dow personnel are available to answer your questions and to provide reasonable technical support. Dow product literature, including safety data sheets, should be consulted prior to use of Dow products. Current safety data sheets are available from Dow.

Medical Applications Policy

NOTICE REGARDING MEDICAL APPLICATION RESTRICTIONS: Dow will not knowingly sell or sample any product or service (“Product”) into any commercial or developmental application that is intended for:

- a. long-term or permanent contact with internal bodily fluids or tissues. “Long-term” is contact which exceeds 72 continuous hours;
- b. use in cardiac prosthetic devices regardless of the length of time involved (“cardiac prosthetic devices” include, but are not limited to, pacemaker leads and devices, artificial hearts, heart valves, intra-aortic balloons and control systems, and ventricular bypass-assisted devices);
- c. use as a critical component in medical devices that support or sustain human life; or
- d. use specifically by pregnant women or in applications designed specifically to promote or interfere with human reproduction.

Dow requests that customers considering use of Dow products in medical applications notify Dow so that appropriate assessments may be conducted.

Dow does not endorse or claim suitability of its products for specific medical applications. It is the responsibility of the medical device or pharmaceutical manufacturer to determine that the Dow product is safe, lawful, and technically suitable for the intended use. **DOW MAKES NO WARRANTIES, EXPRESS OR IMPLIED, CONCERNING THE SUITABILITY OF ANY DOW PRODUCT FOR USE IN MEDICAL APPLICATIONS.**

Disclaimer

NOTICE: No freedom from infringement of any patent owned by Dow or others is to be inferred. Because use conditions and applicable laws may differ from one location to another and may change with time, the Customer is responsible for determining whether products and the information in this document are appropriate for the Customer’s use and for ensuring that the Customer’s workplace and disposal practices are in compliance with applicable laws and other governmental enactments. Dow assumes no obligation or liability for the information in this document. **NO WARRANTIES ARE GIVEN; ALL IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE ARE EXPRESSLY EXCLUDED.**

NOTICE: If products are described as “experimental” or “developmental”: (1) product specifications may not be fully determined; (2) analysis of hazards and caution in handling and use are required; (3) there is greater potential for Dow to change specifications and/or discontinue production; and (4) although Dow may from time to time provide samples of such products, Dow is not obligated to supply or otherwise commercialize such products for any use or application whatsoever.

Additional Information

North America

U.S. & Canada: 1-800-441-4369
1-989-832-1426
Mexico: +1-800-441-4369

Europe & Middle East

All Countries: +31-11567-2626
+800-3694-6367
Italy: +800-783-825

Latin America

Argentina: +54-11-4319-0100
Brazil: +55-11-5188-9000
Colombia: +57-1-219-6000
Mexico: +52-55-5201-4700

Africa

South Africa: +800-99-5078

Asia Pacific

All Countries: +800-7776-7776
+60-7965-5392

<http://www.dow.com>

Published November 2019

© 2019 The Dow Chemical Company